

Monash University
FIT5147 Data Exploration and Visualisation
Semester 1, 2026

Programming Exercise 2: R (5%)

Please carefully review all the requirements below to ensure you have a good understanding of what is required for your assessment.

- 1. Instructions & Brief**
- 2. Assessment Resources**
- 3. Assessment Criteria**
- 4. How to Submit**
- 5. Word Count & Penalties**

1. Instructions and Brief

In this assignment you will demonstrate your capability in creating both a static and interactive visualisation with simple narrative elements using **R**. The exercise expects the use of three R packages: RShiny, ggplot2 and Leaflet as taught in the unit applied activities in Weeks 2-4. For the submission you will be expected to submit all R scripts to run the application (including the data file) and a written reflection piece providing justification and your considerations relating to your visualisation based on the unit's taught theory for data visualisation and visual analytics from Weeks 1-5 inclusive. **It is an individual assignment and worth 5% of your total mark for FIT5147.**

Relevant learning outcomes for FIT5147:

- 1: Perform exploratory data analysis using a range of visualisation tools;
- 3: Critically evaluate and interpret a data visualisation;
- 5: Choose an appropriate data visualisation;
- 6: Implement interactive data visualisations using R and other tools.

Originality:

This is an individual assignment and must be original. Any sources you use to aid in the completion of the assignment must be acknowledged appropriately to ensure the academic integrity of your work. See the assessment page on Moodle for more information, including citing sources. If you have used **Generative AI** you must declare the tool(s) and prompt(s)

used at the end of your reflection document. This includes using any software that paraphrases, translates or rewrites your text or code.

1.1 The Data

The data set used in this assignment is based on the [Atlas of Living Australia \(ALA\)](https://ala.org.au/) online resource. We introduced this dataset in the Week 1 Workshop and Programming Exercise 1 but have prepared a specific data file for this assignment. The PE2 data was extracted from ALA on 24 March, 2026. The data file can be found on Moodle in the Assessments section under the Programming Exercise 2 heading.

To enhance your understanding of the context and metadata, you can check the data source link. Using the various interactive tools for the data source of the full dataset may help enrich your visual analysis, for example the ALA has a map that allows you to explore the data spatially: <https://spatial.ala.org.au/>. If you discuss or replicate the visualisations or metadata provided by ALA, be sure to reference these correctly in your report¹.

The name of the data set is **ALA_S12026PE2.csv**, and described in Table 1. This looks similar to the PE1 data in the table, but it contains different rows and fewer columns. You may further wrangle the data if you wish. In order to complete this task, you may need to add your own columns derived from this subset only. No additional data is required. Wider research to provide context to help you understand your insights may be needed, but is minimal. No data cleaning is expected as part of this exercise. If there is anything you do want to comment on that may hinder the visualisation or the interpretation, do note these in your reflection document.

Specifically, for PE2 we have only included data from 2004-2024 and included data for three additional bird species. Now we have observations for not only the Swift Parrot (*Lathamus discolor*), but also the Little Lorikeet (*Parvipsitta pusilla*), the Orange-bellied Parrot (*Neophema (Neonanodes) chrysogaster*), and the Purple-crowned Lorikeet (*Parvipsitta porphyrocephala*). See Figure 1 for an illustration and Table 1 for the overview of the data:



Figure 1. from left to right: Swift parrot, Purple-crowned lorikeet, Orange-bellied parrot, Little lorikeet.

¹ ALA does include a [citation guide](#), but you may have to adapt the formats they supply to those specified for this unit.

Table 1. Fields of the “ALA_S22026PE2” data set

Column	Description
observationDate	The date of observation or specimen collection date. (Extracted from the EventDate column)
stateProvince	The state or territory matching the supplied coordinates - http://rs.tdwg.org/dwc/terms/stateProvince
decimalLatitude	The decimal latitude associated with this record. If the original latitude was supplied in degrees, minutes, seconds, this value represents the converted value. - http://rs.tdwg.org/dwc/terms/decimalLatitude
decimalLongitude	The decimal longitude associated with this record. If the original longitude was supplied in degrees, minutes, seconds, this value represents the converted value. - http://rs.tdwg.org/dwc/terms/decimalLongitude
scientificName	The name the Atlas has matched this record to in the NSL - http://rs.tdwg.org/dwc/terms/scientificName
vernacularName	The common name the ALA has matched this record to in the NSL - http://rs.tdwg.org/dwc/terms/vernacularName
dataResourceName	The data resource that supplies the record. This is typically a CSV or DwCA file, or a web service.
basisOfRecord	What this is a record of e.g. specimen, human observation, fossil - http://rs.tdwg.org/dwc/terms/basisOfRecord
recordID	Atlas unique ID for this record - http://rs.tdwg.org/dwc/terms/recordID

References: ALA. (n.d.). Atlas of Living Australia (ALA), Retrieved March 24, 2026, from <http://www.ala.org.au>

1.2 The Design Brief

The task is to use R Shiny, ggplot2, and Leaflet to create a **data visualisation page** using the provided dataset and following a specified layout mockup and design requirements.

For this assignment, you will need to create two key visual components: **VIS 1** and **MAP**, combined into one layout.

The mock-up shown in Figure 2 is the expected layout of the visualisation page, and the expected content in each section. In order to create this layout, you must use a **fixedPage** (not a fluidPage) layout and are expected to position the visualisation elements in an appropriate number of rows and columns.

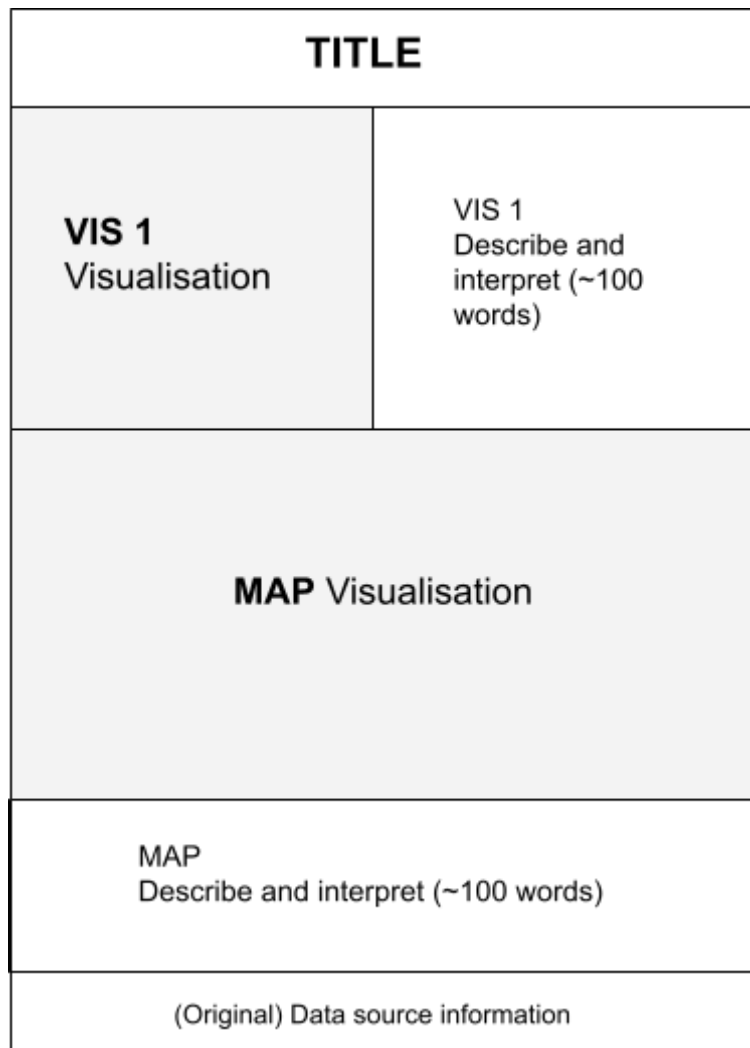


Figure 2. Mock-up showing the approximate position for your two visualisation elements (VIS 1 and MAP) and the two descriptions, title and source. Word counts are approximate.

1.2.1 VIS 1

For this assignment, this visualisation should be **static** (not interactive) with the following requirements:

1. **VIS 1** should show the distribution of seasonal bird observations. **This must show the breakdown of the proportion of observations per season**, using appropriate visualisation idioms and visual variables of your choice. This should allow a fair comparison of the proportions across each of the four Australian seasons.
2. VIS 1 must be created using **ggplot2**.

1.2.2 The MAP

For the **map component**, you need to create an **interactive** dot map or proportional symbol map using *Leaflet*. You must follow the following requirements:

1. For each observation, plot the species on your map using circle markers or equivalent, with the following data mappings and design aspects:

- a. **Each symbol** on the map represents an observation or observations.
 - b. **Colour** should be mapped to the species (you can choose whether you choose to use scientific or common names, or both). You must choose an appropriate colour palette for the type of data.
 - c. **Radius** should be mapped to the number of distinct observations (you may need to scale the size so as to reduce the data overlap on the map, or use opacity, but some overlap is expected).
 - d. Provide a **colour legend** for your map. Ensure the colour choice and ordering of the legend (and layering of the map) is appropriate for the data.
2. Implement the following interactive features:
- a. Provide a **tooltip** for each circle marker on mouse hover-over that shows the scientific and common name of the species, the number of observations at that location and the Australian state or territory.
 - b. Add an interactive **checkboxlist** for filtering which of the bird species are shown on the map. By default, all birds should be shown.
 - c. Add an interactive **radio button list** for filtering five mandatory options including the full year (all data) and one for each **Season**. By default full year should be on.
 - d. By default, the map must be centred on Australia and must include the whole country. You may find these settings useful for initially creating your map (feel free to adapt them for your map's needs and design choices): Centre point of latitude -30.4486, longitude 133.9453, zoom level 3.

1.2.3 The (Data Visualisation) Narrative

- 1) Provide a descriptive **title** for your visualisation.
- 2) The descriptive text in the description boxes should include a brief **description of the** visualisation, guidance on the **interaction, and some interpretation** of the insights shown with the visualisation. They should help the viewer see some data insights you have identified, especially when using the interactive features. The interpretation for the map must discuss species migration pattern(s) across Australia.

- 3) Information on the original (not the file we are using, but the one it is adapted from) **data source** should be provided for all data visualisations². In the relevant layout location (see Figure 1) briefly provide information about the original data source, including the:
 - a) name of the data;
 - b) URL to the data;
 - c) name of the licensor;
 - d) date of the version of the original data used for the visualisation.

1.2.4 Reflection

Using the provided template, write a brief justification and reflection of your completed visualisation based on the visualisation and visual analytics theory you have learnt in the unit so far. This should include justification and any considerations relating to your choice of visualisation, choice of visual variable/s and the visualisation layout, visual hierarchy and typography. As shown in the template, this needs to be completed for each of the two visualisation components: VIS 1 and MAP. The reflection is expected to be text only.

Notes for the Design Brief:

1. You should avoid explicitly hard coding any of the data, where possible.
2. Minimise the amount of data processing done by the server code when it reacts to new user interactions.
3. Consider the user's needs when making design choices. In this case, your audience is University staff with basic or advanced knowledge of Australian geography but limited knowledge of these parrots and their migration patterns.
4. The word counts are a guide. For this textual content we are looking for concise descriptions.
5. The text descriptions are to help your visualisation reader see what you see in the data visualisations, i.e., tell the data story. The reflection is to help your markers understand your design choices using the data visualisation theory and practice taught so far (Weeks 1 to 5).
6. The design choices for VIS 1 and MAP will be included in the evaluation of your work. Make sure you meet the requirements of the assignment, choose any visual elements wisely and try to ensure each component of the layout, including typography, can be justified.
7. Whilst a map legend is usually expected for all map elements, a legend for the size of the proportional symbols on the map is not a requirement for this assignment. (You may provide one if you wish for completeness.)
8. No data checking or cleaning is required, but you may need to perform limited data wrangling. You could use an R package such as *dplyr* (<https://dplyr.tidyverse.org/>) for this purpose, but it is not a requirement.

² Providing the data source allows users of your visualisation to find and access the original data themselves and allows them to better understand the original purpose/ownership of the data.

9. Whilst there are no requirements on the use of specific colour palettes for this assignment, appropriate use of colour is essential and choice of visual variable should be justified. We recommend the use of *ColorBrewer* palettes: (https://ggplot2.tidyverse.org/reference/scale_brewer.html).
10. The text for your description text boxes, titles and any visualisation element content should be easy to read. Choose a clear font and appropriate font size for your submission and be sure to check your grammar and spelling.
11. Your written reflection and code is run through academic integrity checks. No collusion between students is permitted and any text or R code that is largely based on any third party code **must cite the original source** in comments within the R scripts(s) (or as a reference, or footnote in your reflection), including webpages or social media messages. Otherwise your work may be considered to be plagiarising the code of others.
12. Any use of tools with generative AI **must be declared at the end of your work**, including identifying the prompt(s) used and their purpose. This includes using such tools for writing, translating, coding, images or any other function.

2. Assessment Resources

See the Assessments section on Moodle for the:

- Data: *ALA_S12026PE2.csv*
- Reflection Template: *PE2_reflection_template_S12026.docx*

3. Assessment Criteria

The following outlines the criteria which you will be assessed against.

- Demonstrate the ability to read in and manage data effectively using R [0.5%]
- Demonstrate the ability to create appropriate static visualisations in R using ggplot2 [1%]
- Demonstrate the ability to create an appropriate data map in R with Leaflet [1%]
- Demonstrate the ability to create an appropriate interactive narrative visualisation in R with Shiny [1.5%]
- Demonstrate the ability to justify and critically reflect on interactive and static data visualisations [1%]

4. How to Submit

Submit two files:

- **A PDF file** containing the justification and reflection. This reflection must use the template provided. Name the PDF file in this format:
PE2_[LAST NAME]_[STUDENT ID].pdf.
- **A ZIP file** containing all files required to run your Shiny application. Name the ZIP file in this format:
PE2_[LAST NAME]_[STUDENT ID].zip.

Before submitting your assignment, double-check that your Shiny application runs correctly. To do so, clear objects from the workspace by clicking on the “Broomstick” icon on the top-right section of RStudio. Afterwards, make sure your application is still working by clicking the “Run App” button on RStudio.

The files that you need to include in your ZIP submission are:

- The **one dataset** supplied for this assignment
- **R script(s)** for the final Shiny application (you can use a single R script, or two scripts for UI and Server)
 - Have all required "library(xxx)" or "require(xxx)" statements at the beginning of your R files (you do not need the code to install the packages)
 - Use relative paths when reading your dataset (do not use absolute paths that refer to specific drives)

Do not include your PDF file in your ZIP file.

5. Word Count and Penalties

The reflective report is expected to have approximately 100 to 200 words per section, totaling approximately 500 words. The supplied template must be used.

As per Monash policy: All late submissions will receive a penalty of 5% per day (0.25 marks per day out of a total of 5 marks) late inclusive, including weekends. Work submitted more than seven days after the due date will not be marked.